

UX UI Intern @ Treevah

Integrating AI into a Folder Management System

Brandon Wong

Treevah : File Management + AI Integration

Streamlined file and folder analysis + organizational insights

- Designer on **cross-functional team**
- Crafted **Desktop Dashboard + AI Experience**
- Conducted **user research**
- Presented to **Treevah CEO**

My role

UX Designer

2

weeks

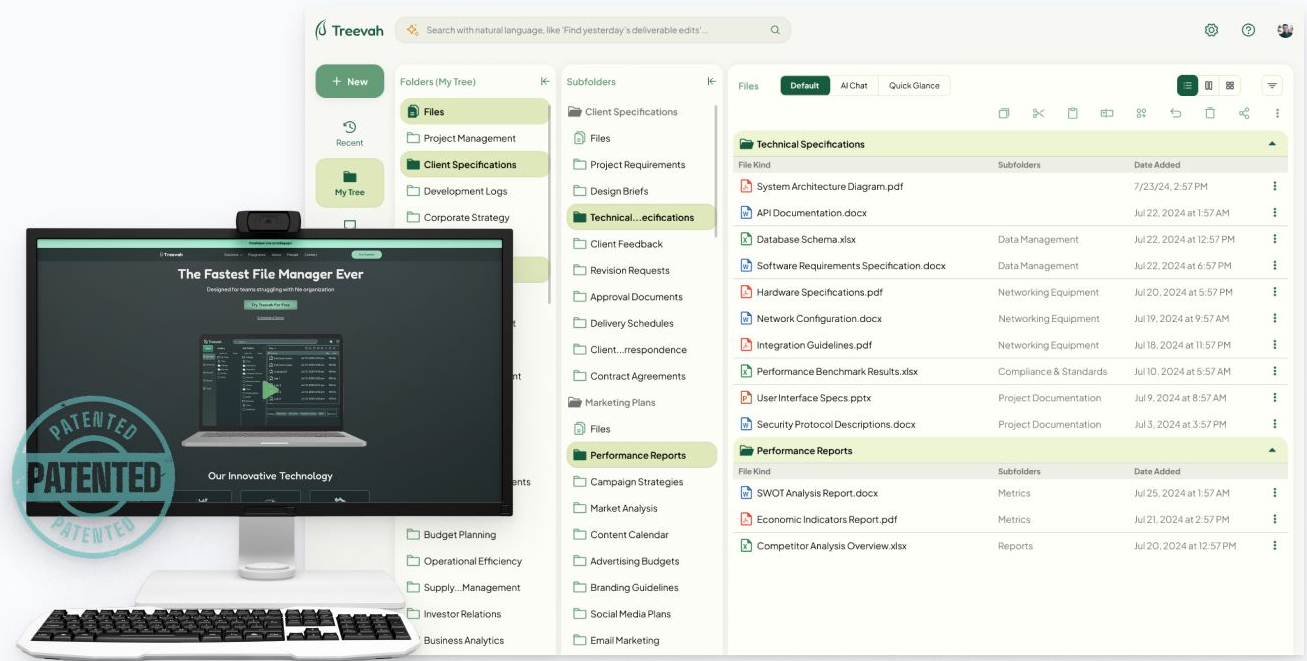
My team

1 PM

1 UX Designer

1 Researcher

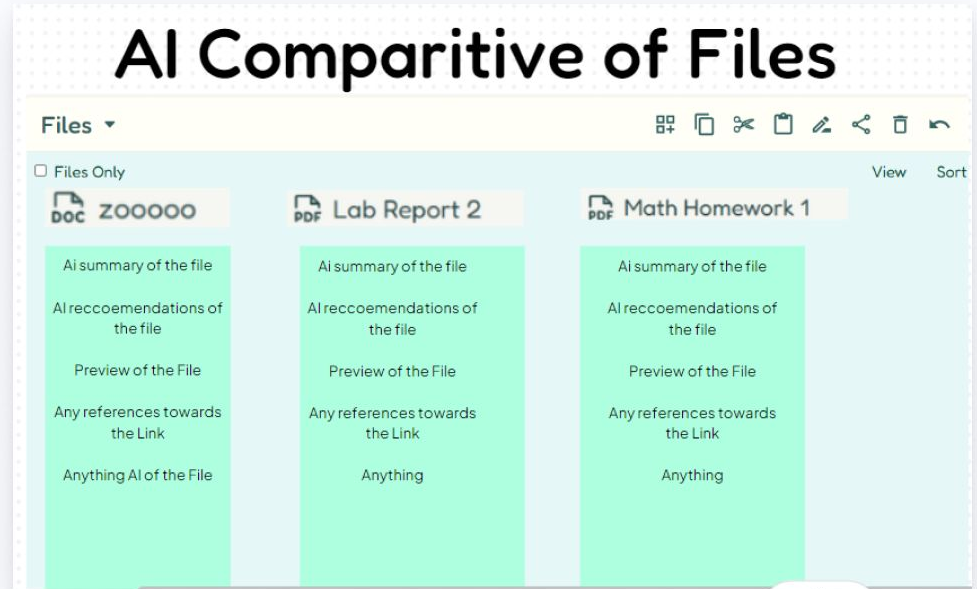
What's Treevah?



Research

Began a 1-week research sprint

- 6 user interviews
- Analyzed Treevah's file management system as well as Google Drive
- Outlined CEO needs to understand projects goals



*CEO's Inspiration Photos

Research

Interview Insights

- Interviewees were concerned about privacy and having AI handle their file.
- Interviewees expressed frustrations when navigating files in nested folder systems
- Understanding file content quickly and efficiently was highlighted

AI functionality AI user	Feedback (visual/assistive) AI user	Quick File Glance AI user	Comparison with desktop AI user	Thoughts on AI AI user
D7: did not feel it was necessary to use AI for the comparison of the two files	D7/B: Figma prototype hints gave away some of the interactions	S2: found Quick File Glance by accident (moved to click on the thumbnail, confused at pop-up suddenly appearing)	L2: liked the AI better than desktop, feels more professional and more evolved, more secure	D7: never seeks assistance from AI for productivity
D7: to try to complete the compare task, clicked top to bottom, tried clicking copy icon, clicked everything in toolbar	D8: did not like that there were buttons they could not press	L2: clicked on multiple other icons and buttons before finding the button. Noticed off the beat on the Figma prototype	D7: liked the desktop prototype, enjoys the idea of being able to see multiple subfolders at once	L2: has concerns about accuracy and information privacy, might only upload certain academic files to ensure privacy
S2: assumed AI feature would allow them to be able to find files easier	L1: completed comparing tasks but doesn't know what was going on, wanted more visual feedback	L2: initially tried clicking on the file preview instead, initially confused about the difference between the file preview and the quick glance	S2: thought desktop interface was straightforward	S2: liked the concept of the AI Quick File Glance feature
S2: thinks the AI feature should be easier to find, "nothing said to use it"	D7: A button, not clear what the file means, expected a mouse over tool tip	L2: in early tried clicking on the file preview instead, initially confused about the difference between the file preview and the quick glance	D1: liked how multiple folders can be open at a time, found to be an easily understood layout	L2: if everything works correctly and the AI is actually reliable, then uses the potential
D7: What do you think is different about this versus other file managers? Was an AI that is not very obvious to access	L1: Thought visual feedback would be helpful and new user might need assistance	D7: opened toolbar and clicked the preview. As before correctly clicking quick file glance	D1: in the desktop, liked tagging of their folders on	D7: thought it in file manager is fine since it will be online, probably won't use it, but once in a while it will be of some help
L1: "AI that we need are how results, better to just provide the tools to find the files themselves"	D1: One of our biggest ones very confused and after the fact maybe would have been more clear about what they're to know (what works and doesn't)	D7: clicked query bar repeatedly for about 30 sec before clicking in-depth analysis	D7: liked the tool for better way of viewing files that usual, also thought desktop was able to be categorized by more filters or something and thought that was a good feature	S2: thinks AI feature for a file manager is okay, a unique concern in the scenario the user has confidentiality that
D8: to compare files, tried clicking new, the comparison pop-up, the files themselves		S2: understood Quick File Glance to have additional information about the file	D7: liked the tool for better way of viewing files that usual, also thought desktop was able to be categorized by more filters or something and thought that was a good feature	S2: does not use AI for productivity at all
D8: thought that maybe if the files were both highlighted and grouped new maybe they would open them up into a new document		S2: did not understand Quick File Glance to be AI	S1: I got confused with the desktop, and didn't understand the functionality. Thought desktop is easier but was intuitive than AI prototype. AI is a little overwhelming	D7: liked the concept of AI being used to see a quick glance of the file
		L1: liked the layout for quick file glance but wishes the feature to also be found with a right click	D7: likes icons too big in the grid view	D7: AI for productivity → uses AI to brainstorm essay topics → questions

“

I would use AI features if I had a few files that were similar and wanted to know which was what.

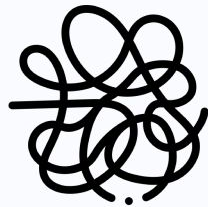
- Interviewee

The Problem

Frustrating File/Folder Analysis. Not enough efficiency.



1. Complex files, lack understanding



2. Disorganized content



3. File privacy

The Objective

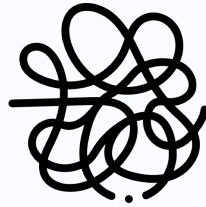
How can we streamline content analysis experience?



1. Complex files, lack understanding



**HMW improve content
analysis of complex files**



2. Disorganized content



**HMW promote organization
of file and folders**



3. File privacy



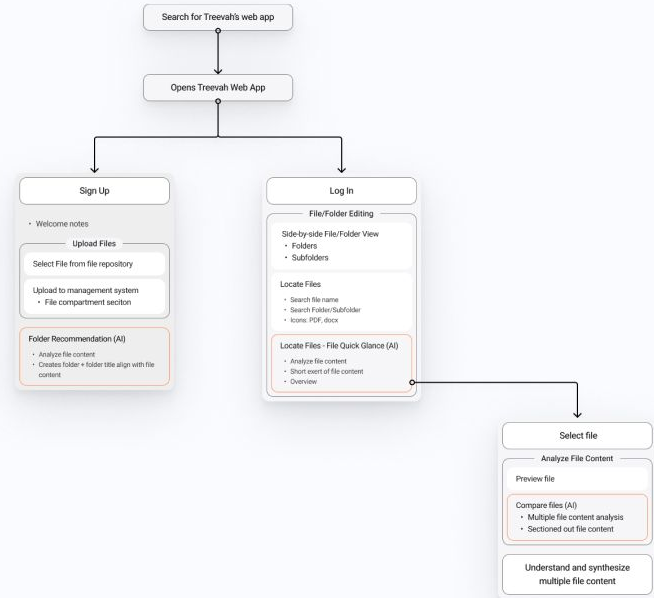
**HMW ensure users are
comfortable with AI use**

User Flow

Typical User flow for Treevah's File Management system

First time user: Sign up > Upload Files > Sort Files > Organized Files into Folders

Return User : Log in > Select Folders > Locate Files > Analyze file

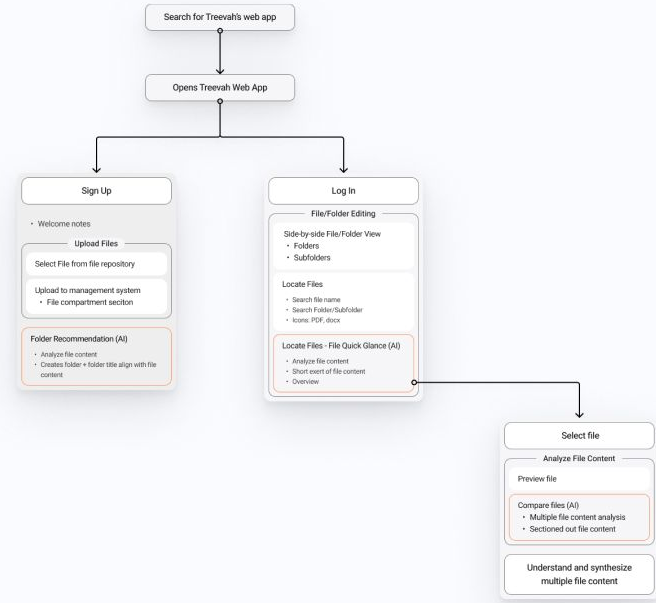


User Flow

Typical User flow for Treevah's File Management system

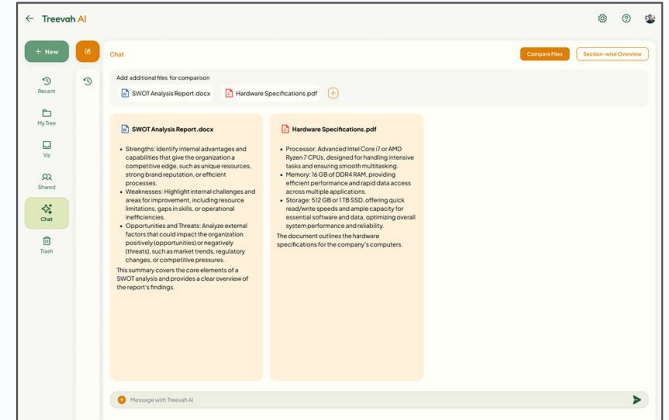
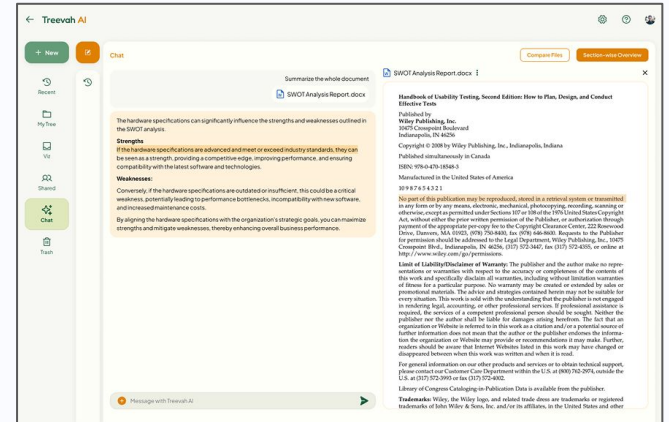
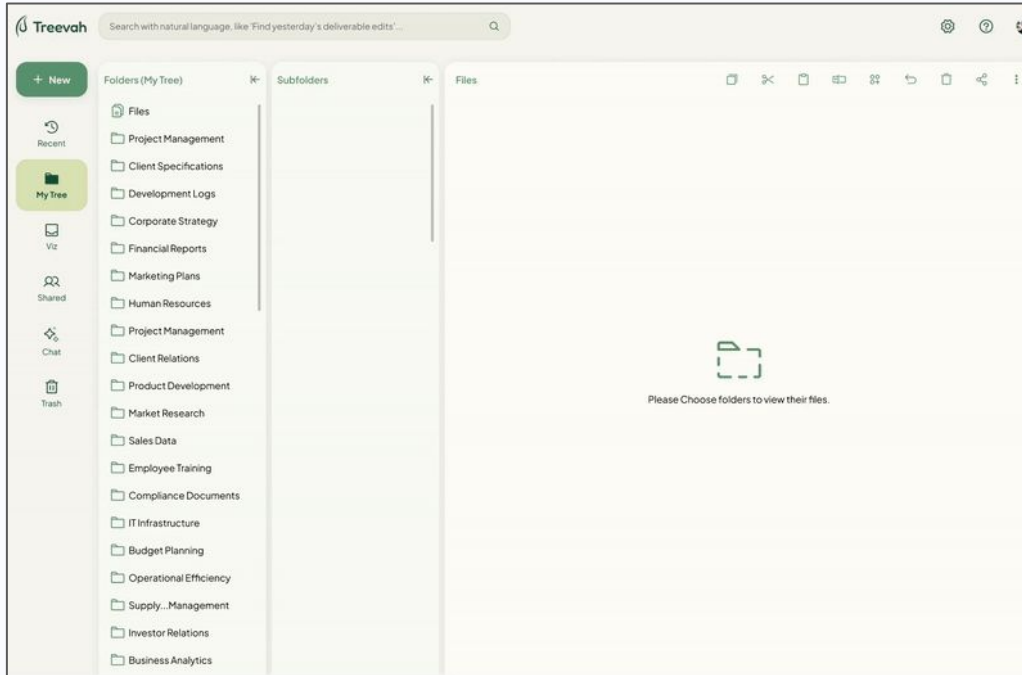
First time user: Sign up > Upload Files > Sort
Files + **Folder Recommendations** > Organized
Files into Folders

Return User : Log in > Select Folders >
Quick File Glance + Locate Files > Analyze files
+ **Comparing File Content**



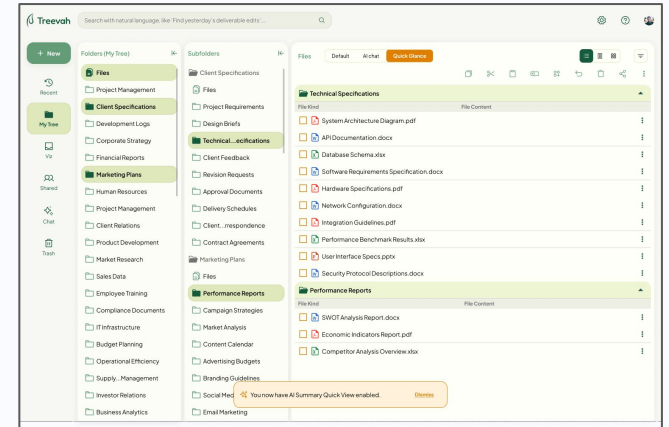
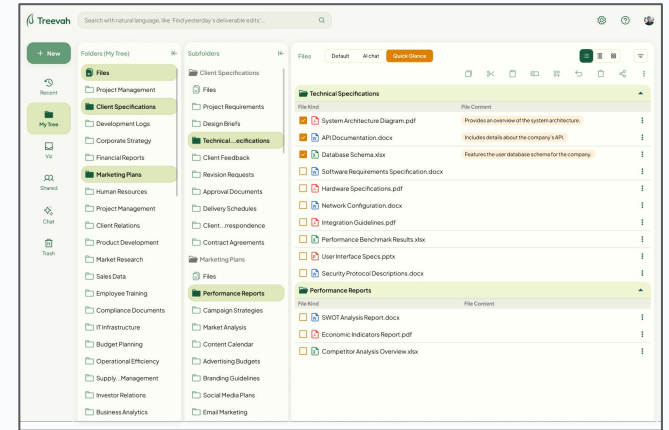
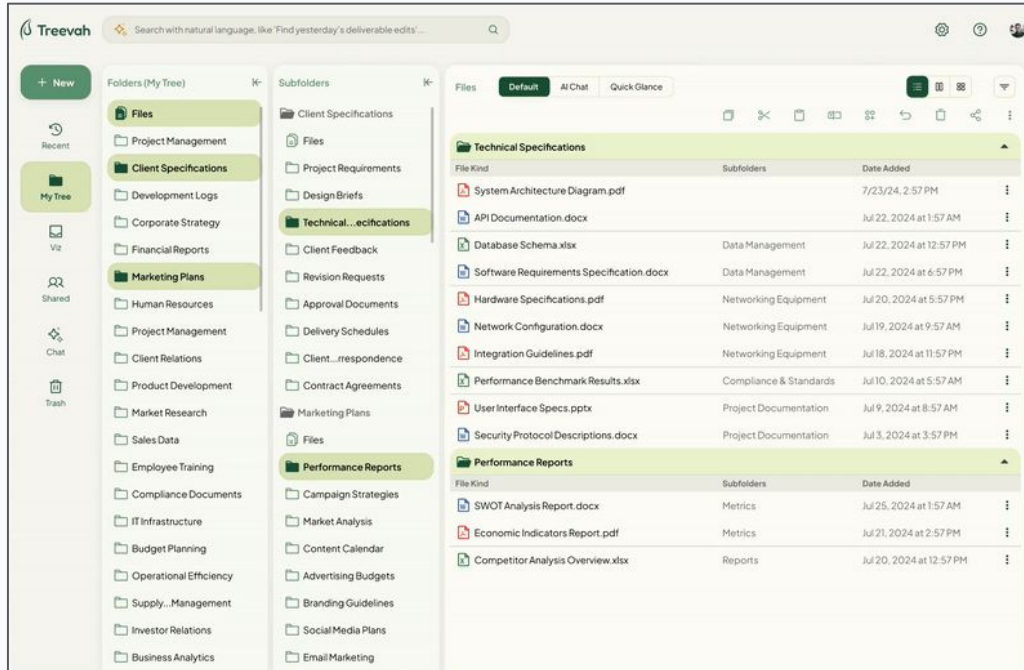
Comparing File Content

Design Solution



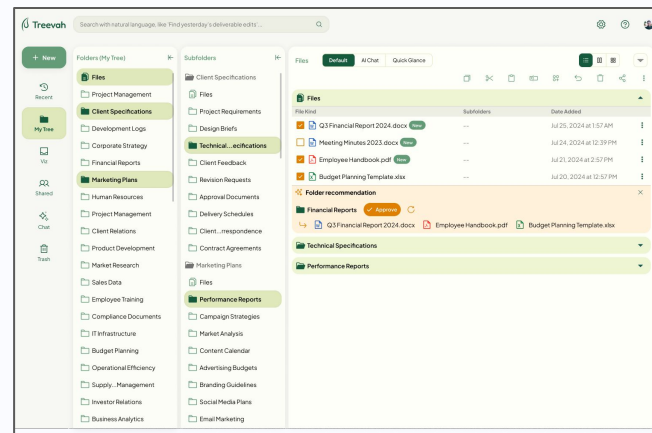
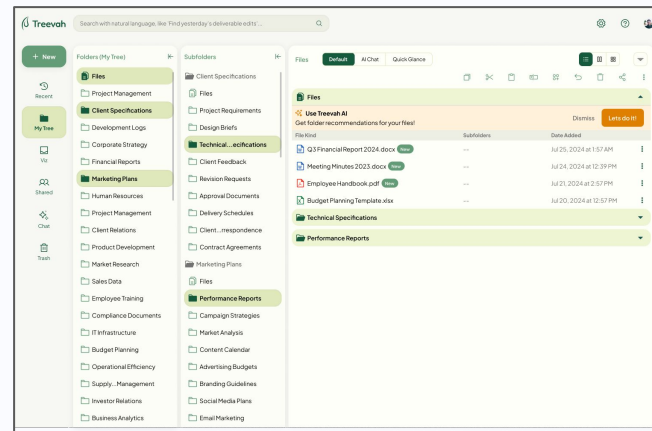
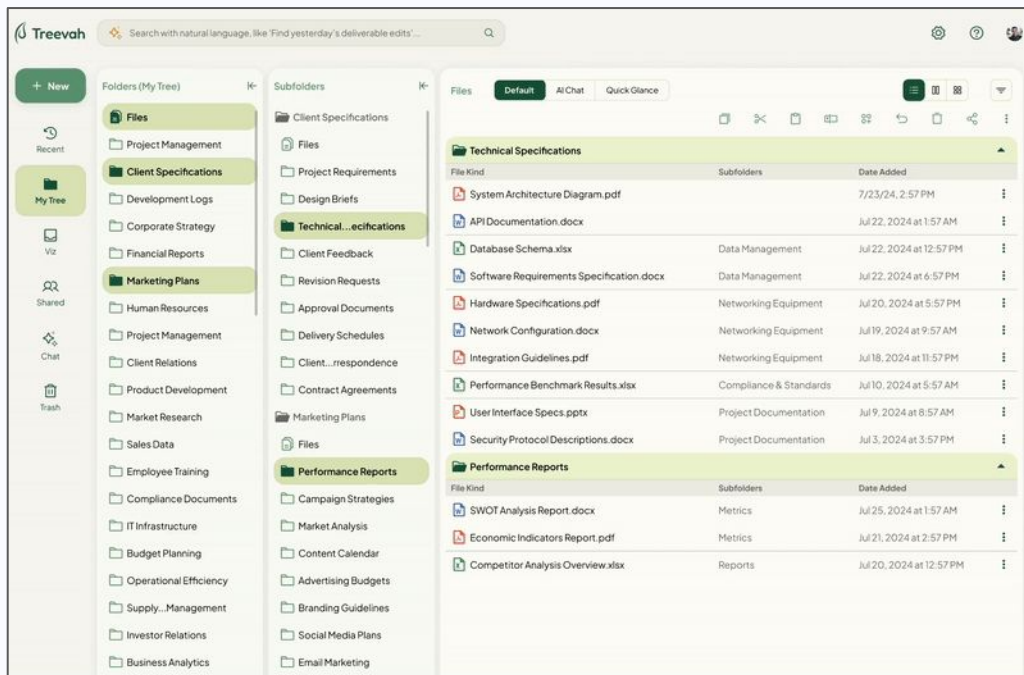
Quick File Glance

Design Solution



Folder Recommendations

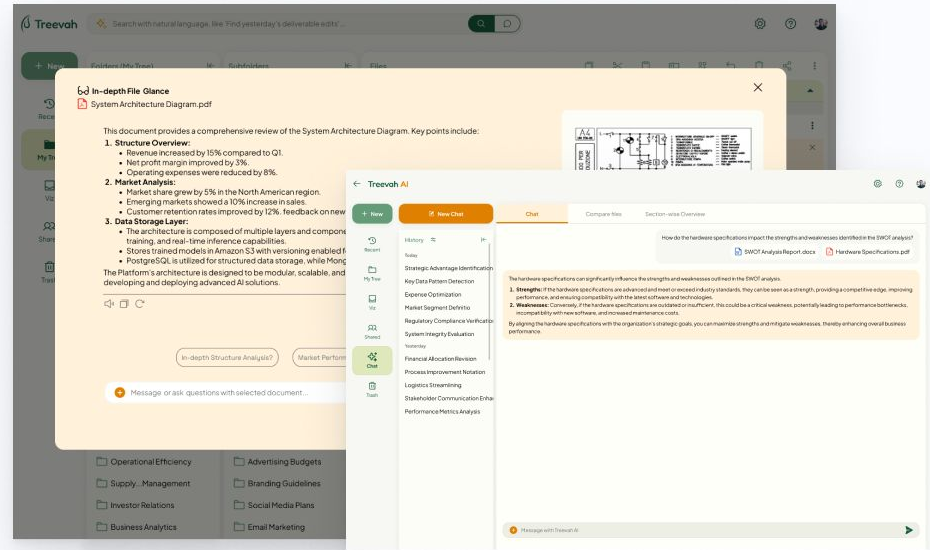
Design Solution



Usability Testing

8 user interviews later...

- Participants did not know where to activate AI.
- AI Functionality lacked visual cues
- Participants did not find the in-depth file analysis useful



Search with natural language, like 'Find yesterday's deliverable edits' ...

Before turning on AI Mode

Choose the files and folders you want to inquire about

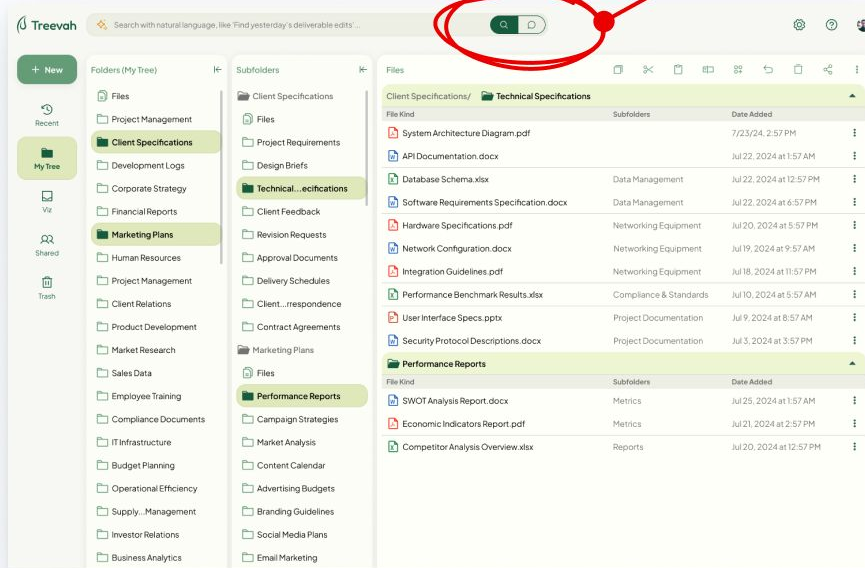
AI Mode turned on

Usability Testing

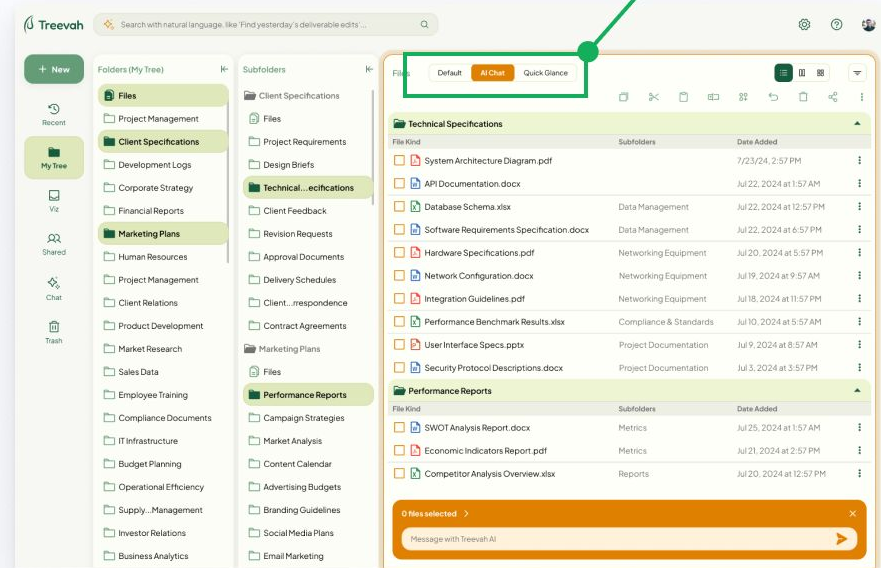
Search Bar

1 of 8 participants recognized the toggle as the activation method for Treevah AI.

7 of 8 participants were able to activate AI. Separated the search bar and AI activation to a separate toggle component.



Old ❌



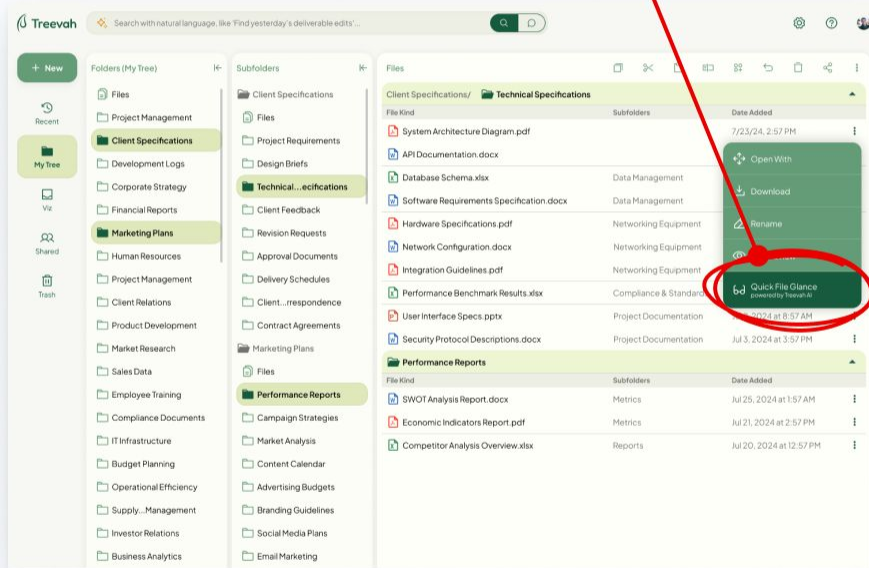
Redesigned ✅

Usability Testing

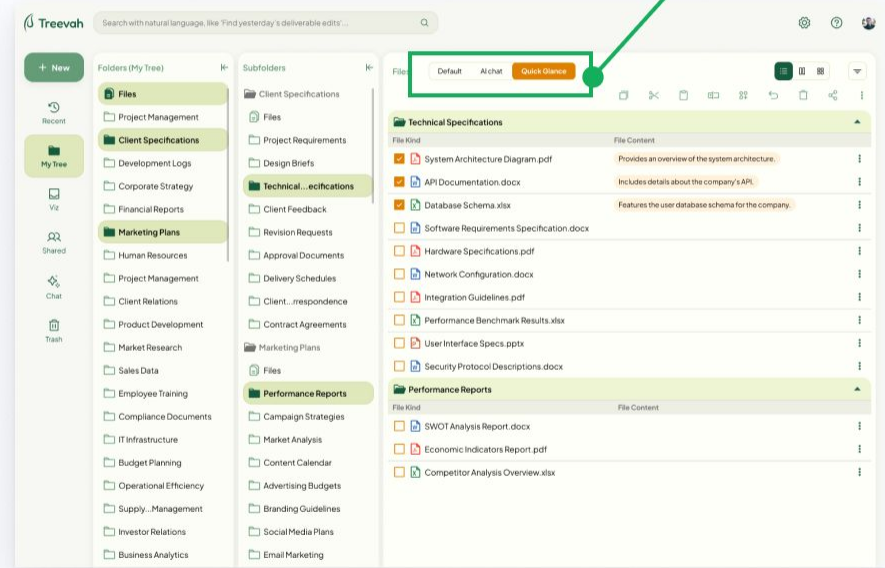
Quick Glance

Only 2 of 8 of participants were able to locate the Quick File Glance feature

100% of participants located Quick Glance



Old ❌



Redesigned ✅

Transparency and Control



Incorporated user choice to build trust and ease AI concerns.

Files

Use Treevoh AI
Get folder recommendations for your files!

Dismiss Lets do it!

File Kind Subfolders Date Added

Treevoh Search with natural language, like "Find yesterday's deliverable edits"...

+ New

Folders (My Tree)

- Files
- Project Management
- Client Specifications
- Development Logs
- Corporate Strategy
- Financial Reports
- Marketing Plans
- Human Resources
- Project Management
- Client Relations
- Product Development
- Market Research
- Sales Data
- Employee Training
- Compliance Documents
- IT Infrastructure
- Budget Planning
- Supply...Management
- Investor Relations
- Business Analytics

Subfolders

- Client Specifications
- Files
- Project Requirements
- Design Briefs
- Technical...ecifications
- Client Feedback
- Revision Requests
- Approval Documents
- Delivery Schedules
- Client...respondence
- Contract Agreements
- Marketing Plans
- Files
- Performance Reports
- Campaign Strategies
- Market Analysis
- Content Calendar
- Advertising Budgets
- Branding Guidelines
- Social Media Plans
- Email Marketing

Files Default AI Chat Quick Glance

File Kind	Subfolders	Date Added
System Architecture Diagram.pdf		7/23/24, 2:57 PM
API Documentation.docx		Jul 22, 2024 at 1:57 AM
Database Schema.xlsx	Data Management	Jul 22, 2024 at 12:57 PM
Software Requirements Specification.docx	Data Management	Jul 22, 2024 at 6:57 PM
Hardware Specifications.pdf	Networking Equipment	Jul 20, 2024 at 5:57 PM
Network Configuration.docx	Networking Equipment	Jul 19, 2024 at 9:57 AM
Integration Guidelines.pdf	Networking Equipment	Jul 18, 2024 at 11:57 PM
Performance Benchmark Results.xlsx	Compliance & Standards	Jul 10, 2024 at 5:57 AM
User Interface Specs.pptx	Project Documentation	Jul 9, 2024 at 8:57 AM
Security Protocol Descriptions.docx	Project Documentation	Jul 3, 2024 at 3:57 PM

Performance Reports

File Kind	Subfolders	Date Added
SWOT Analysis Report.docx	Metrics	Jul 25, 2024 at 1:57 AM
Economic Indicators Report.pdf	Metrics	Jul 21, 2024 at 2:57 PM
Competitor Analysis Overview.xlsx	Reports	Jul 20, 2024 at 12:57 PM

2 files selected > Clear X

Message with Treevoh AI

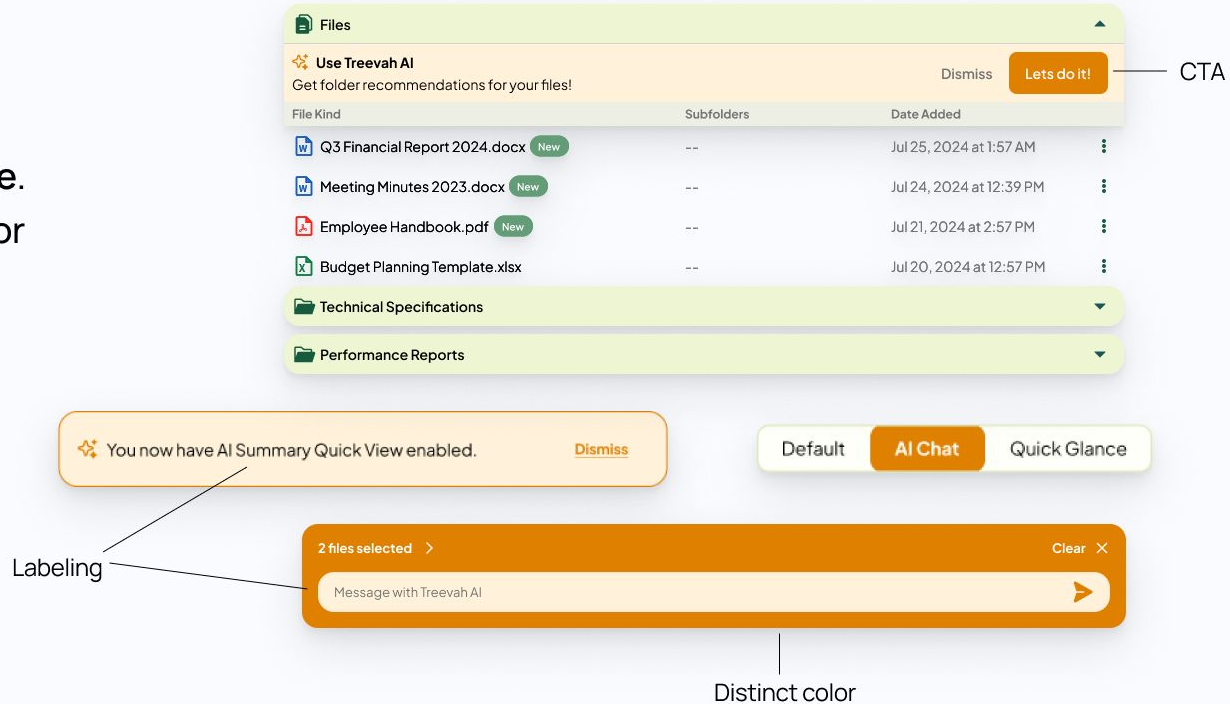
Opt-in

Progress visibility

User-initiated

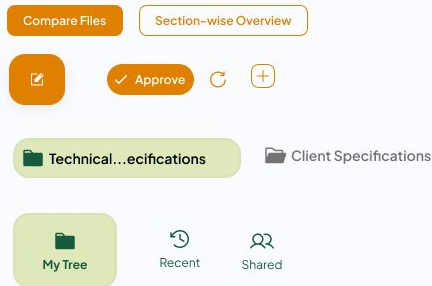
Feature Segmentation

Treevah AI is a premium feature.
Used **color** and **components** for
visual differentiation.



Why Icons?

Humans are inherently **visual beings**.

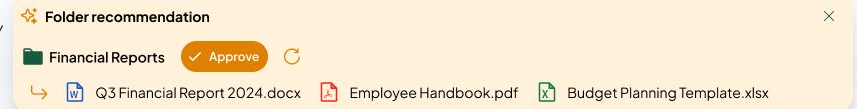


Indication of
a selected file

Visualizable file types

Indicating AI

Technical Specifications			
File Kind	Subfolders	Date Added	
System Architecture Diagram.pdf		7/23/24, 2:57 PM	⋮
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User Interface Specs.pptx	Project Documentation	Jul 9, 2024 at 8:57 AM	⋮
Security Protocol Descriptions.docx	Project Documentation	Jul 3, 2024 at 3:57 PM	⋮

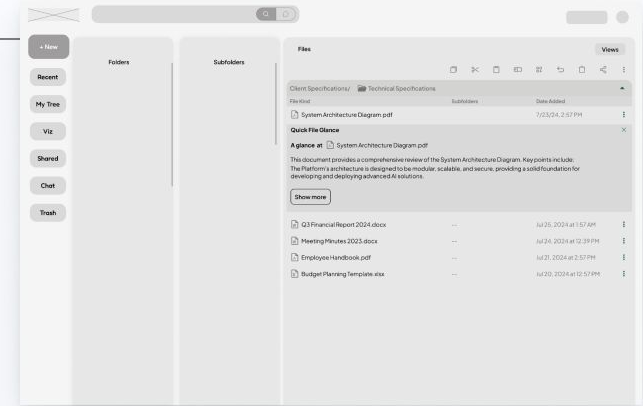


🌟 You now have AI Summary Quick View enabled. Dismiss

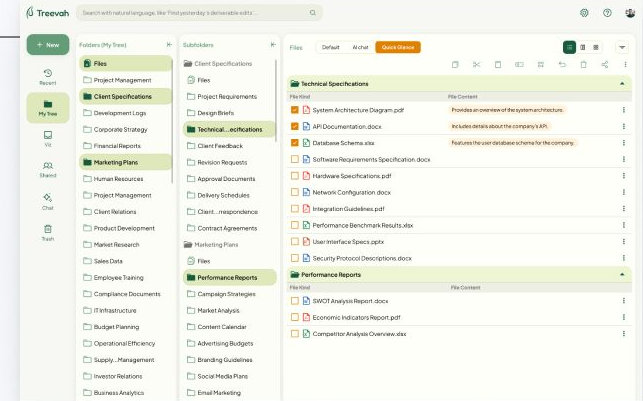
Overcoming Constraints

- Curated feature prioritization list
- Design and test in lo-fi

Low fidelity



High fidelity



User Feedback informed design iterations

Embrace flexibility

- Tailored understanding
- Be agile
- Guide stakeholder expectations with research insights

